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Method for compensating multi-time measurement data of distribution network based on alternating minimization matrix completion combined with VMD-ARIMA-LSTM

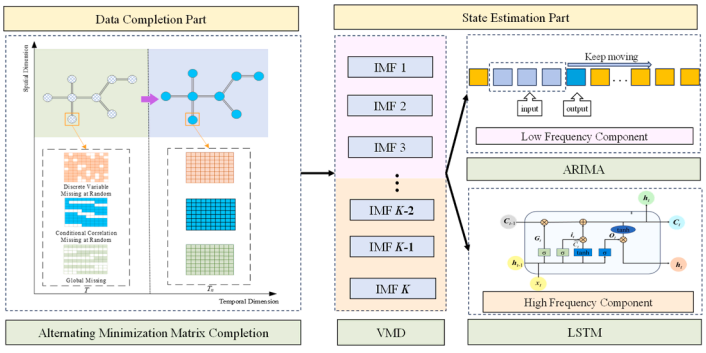
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HIGHLIGHTS

- According to the characteristics of the measured data of distribution network, we develop a physics-informed matrix completion framework that incorporates multiphase power flow equality constraints, with hybrid data-physics fusion strategy enhanced missing-data recovery capability.
- The algorithms in this paper also consider the missing of single section data and time series data, improves the data scale that can be completed by the algorithm, and controls the completion error at a low level, ensuring the effectiveness of the algorithm in the analysis of distribution network measurement data.
- The proposed methodology employs VMD to decouple the inherent multi-scale characteristics of spatiotemporal measurement data, where the low-frequency components are processed through ARIMA modeling for trend extrapolation, while high-frequency residuals are analyzed via LSTM networks to capture temporal dynamics. This hybrid decomposition-reconstruction architecture demonstrates prediction accuracy improvements of 38 % RMSE reduction.

GRAPHICAL ABSTRACT



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- Under severe data incompleteness conditions, the proposed algorithm maintains solution stability through adaptive regularization based on real-time observability evaluation and physical constraint, effectively prevents significant estimation bias or even non-operation situations.

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ABSTRACT

Modern distribution networks with high penetration of distributed energy resources (DERs) are undergoing continuous expansion in scale. However, the increasing complexity of network structure and the high installation cost of measurement equipment introduce operational challenges including state variability and measurement data incompleteness. Substantial data loss significantly compromises fault detection accuracy and network performance, creating obstacles for distributed energy management and posing critical challenges to distribution network state estimation. To address these issues, this paper proposes a hybrid state estimation framework (MC-VMD-ARIMA-LSTM) that integrates alternating-minimization matrix completion (MC) with variational mode decomposition (VMD), autoregressive integrated moving average (ARIMA) modeling, and long short-term memory (LSTM) neural networks for enhanced power flow analysis in low-observability distribution networks. The methodology features a dual-timescale approach: (1) At individual time intervals, an alternating-minimization matrix completion model is formulated, incorporating linearized power flow constraints; (2) For multi-timescale analysis, the measurement dataset undergoes VMD-based decomposition, with subsequent specialized processing where ARIMA handles low-frequency components and LSTM manage high-frequency residuals. The results of state estimation are obtained through systematic component reconstruction. Comprehensive evaluations using IEEE 33-bus distribution network and actual distribution system measurement datasets demonstrate the framework's effectiveness in both multi-timescale data assimilation and state estimation accuracy under limited observability conditions.

1. Introduction

Aligned with global energy transition objectives, significant efforts are being devoted to developing advanced power systems with enhanced sustainability characteristics [1]. Distribution network is evolving into a pivotal area within next-generation power systems due to their critical role in energy provisioning. This transformation drives integrated coordination of generation-grid-load-storage systems, where the proliferation of distributed energy resources (DERs) like photovoltaic systems and the emergence of flexible loads like electric vehicles (EVs) have accelerated the digitalization process of distribution network construction.

Compared to traditional distribution networks, modern distribution networks incorporating DERs and EVs demonstrate distinct stochastic operational characteristics with inherent intermittency and variability. While this technological evolution has augmented power delivery capabilities and introduced dynamic operational flexibility, it substantially elevates system uncertainty during real-time dispatch. These developments drive evolving measurement topologies, altered load patterns, increased fault susceptibility, and exponential growth in monitoring data volume. Such conditions heighten risks of data integrity loss, encompassing transmission errors, communication-induced failures, and anomalous data occurrences. Furthermore, infrastructure deployment constraints involving measurement device installation costs and technical complexity hinder comprehensive monitoring coverage in rapidly evolving networks. Physical degradation mechanisms including equipment aging and external damage compound these operational challenges [2]. Therefore, this paper focuses on the study of data completion techniques for distribution networks under conditions of limited measurements to support precise and efficient control and management methods for distribution networks.

Single-time-slice data missing represents a prevalent challenge in distribution network operational monitoring, particularly affecting real-time state estimation accuracy. Weighted least squares method (WLS) [3] is widely used in state estimation of power systems. Due to insufficient redundancy in the distribution network measurement, it is difficult for WLS to provide a credible assessment of the system operating status [4]. Current approaches to enhance system observability primarily focus on two strategies: (1) optimal sensor placement optimization [5,6] and (2) pseudo-measurement generation algorithms [7,8]. Nevertheless, practical deployment is constrained by sensor deployment economic viability limitations and inherent pseudo-measurement estimation inaccuracies [8,9], ultimately leading to compromised solution feasibility. Ref [10] used neural networks for state estimation, which can ensure the estimation accuracy remains within a reasonable range during communication failures. Ref [11] developed a compressive sensing (CS) framework leveraging inter-measurement linear correlations for state estimation of three-phase unbalanced distribution network. Ref [12] formulated a robust matrix completion paradigm with inherent noise immunity, specifically designed for poor data quality conditions.

However, existing completion techniques fail to address temporal dependencies inherent in time-series data. Significant research efforts have been devoted to developing missing-data resilient state estimation algorithms tailored for modern distribution networks. A new tracking state estimator is proposed in [13] to perform distribution network state estimation by combining PMU and SCADA measurements at the same time with the predicted values obtained from previous state estimation information. Ref [14] established a tensor completion paradigm to fill in the missing data in the tensor by using the spatiotemporal correlation of the measurements. According to randomly generated incomplete data sets (10 % missing rate), ref [15] selected and generated typical industrial-commercial load profiles, and used linear interpolation, spline interpolation and Expectation-Maximization (EM) interpolation methods to complete the missing data. FENG et al. Ref [16] developed a grey adaptive K-nearest neighbor (GAKNN) method, which first constructs time series features, then sets threshold values to screen nearest neighbor points on the basis of naive K-nearest neighbor (KNN) method, calculates the weight coefficient of nearest neighbor points combined with grey association coefficient, finally completes the missing data in turn.

Nomenclature

Indices

M	Real-valued data matrix
Ω	Known elements in real-valued data matrix
$\boldsymbol{X}$	Completed matrix
$\mathscr{B}$	A set of buses
$\mathscr{L}$	A collection of branches of distribution network
f	Begin of the branch
t	End of the branch
m	Number of rows in the matrix
n	Number of columns in the matrix
$\Re(\cdot)$	Real part of the complex variable
$\Im(\cdot)$	Imaginary part of the complex variable
ε	Bus number
w	Phasor of the no-load voltage of the non-slack bus
r	Rank of matrix
U, V	Matrixes split from X
η	Noise sequence of ARIMA equation
f(T)	Original signal to be decomposed
$u_{K(T)}$	Input signal of VMD
$A_{K(T)}$	Amplitude
$\phi_{K(T)}$	Phase
$\delta_K(T)$	Dirac function
ω	Central frequency
x_q	Actual value of the qth data point
$\widehat{x}_q$	Estimated value of the qth data point
$\xi_{r,ft}$	Error tolerance of the real part of Ohm's Law
$\xi_{c,ft}$	Error tolerance of the complex part of Ohm's Law

Parameters and constants

δ	Degree of measurement noise
$\mathbf{Y}$	Bus admittance matrix
$\mathbf{y}_{ft}$	Line admittance
$\boldsymbol{B}$, $\boldsymbol{C}$	Parameters of linearized power flow equation
μ, ρ	Penalty parameters of matrix completion model
z, c	Constants of matrix completion model
α	Residual threshold
φ	Autoregressive coefficient
$\tau_r,\tau_c,\gamma,\alpha_r,\alpha_c$	Error tolerances of noise adaptability constraints

Variables

i	row number of a data point in the matrix
j	column number of a data point in the matrix
$\mathbf{i}_{ft}$	Current in each branch
$\mathbf{v}$	Bus voltage
$\mathbf{s}$	Bus injection power
k	Number of iterations when solving the matrix completion model
$\boldsymbol{p}$	Autoregressive orders
$\boldsymbol{q}$	Moving average orders
$\boldsymbol{K}$	Number of VMD modes

Acronyms

MC	Matrix completion
VMD	Variational mode decomposition
ARIMA	Autoregressive integrated moving average
LSTM	Long short-term memory
DERs	Distributed energy resources
EVs	Electric vehicles
WLS	Weighted least squares
SCADA	Supervised control and data acquisition
AMI	Advanced metrology infrastructure
PMU	Phasor measurement unit
MA	Moving average
AR	Autoregressive
IMFs	Intrinsic mode functions
RMSE	Root Mean Square Error
MSE	Mean Squared Error
MAPE	Mean Absolute Percentage Error
R^2	R-Square
CS	Compressive sensing
EM	Expectation-maximization
GAKNN	K-nearest neighbor
KNN	K-nearest neighbor

While conventional matrix completion (MC) techniques have demonstrated effectiveness in completing missing data, their deployment in power system reveals two critical limitations: (1) inadequate adaptation to the spatiotemporal coupling characteristics inherent in power system datasets. (2) inability to achieve concurrent processing of long temporal sequences and single-time scale state estimation. Moreover, the existing distribution system state estimation (DSSE) algorithms still rely on the measurement data of measuring equipment, and the distribution network's situation cannot be reliably estimated when the proportion of equivalent measurement data is missing is extremely high. Therefore, it is necessary to study algorithms that can reconstruct complete network states only from currently available data at low observable levels.

This paper proposes a matrix completion algorithm based on alternating minimization for low measurement conditions of distribution systems, which can only use single-time-instant measurements to reconstruct missing data points at the same temporal resolution. Building upon this foundation, we develop the MC-ARIMA hybrid architecture that achieves unobserved bus data recovery through spatiotemporal coupling analysis, enabling ultra-short-term distribution system forecasting. To address large-scale spatiotemporal data missing common in modern active distribution networks, this paper further proposes MC-LSTM and its improved algorithm MC-VMD-ARIMA-LSTM. Experimental validation demonstrates our models can handle simultaneous spatial-temporal data losses exceeding 50 %. The main contributions of this paper are as follows:

- According to the characteristics of the measured data of distribution network, we develop a physics-informed matrix completion framework that incorporates multiphase power flow equality constraints, with hybrid data-physics fusion strategy enhanced missing-data recovery capability.
- The algorithms in this paper also consider the missing of single section data and time series data, improves the data scale that can be completed by the algorithm, and controls the completion error at a low level, ensuring the effectiveness of the algorithm in the analysis of distribution network measurement data.
- The proposed methodology employs VMD to decouple the inherent multiscale characteristics of spatiotemporal measurement data, where the low-frequency components are processed through ARIMA modeling for trend extrapolation, while high-frequency residuals are analyzed via LSTM networks to capture temporal dynamics. This hybrid decomposition-reconstruction architecture demonstrates prediction accuracy improvements of 38 % RMSE reduction.
- Under severe data incompleteness conditions, the proposed algorithm maintains solution stability through adaptive regularization based on real-time observability evaluation and physical constraint,

effectively prevents significant estimation bias or even non-operation situations.

2. Matrix completion Model

Assume there have a low-rank incomplete matrix, the goal of the matrix completion is to determine unknown elements in that matrix. Let $\boldsymbol{M} \in \mathbb{R}^{m\times n}$ be a real-valued data matrix, set $\boldsymbol{\Omega}$ as known elements in it, $\Omega \subseteq \{1,\ldots\ldots m\} \times \{1,\ldots\ldots n\}$, and $\boldsymbol{M}_\Omega \in \mathbb{R}^{m\times n}$ represents a data matrix containing known measurements, among which $(\boldsymbol{M}_\Omega)_{i,j} = \boldsymbol{M}_{i,j}$ and $(i, j) \in \Omega$, the rest parts of $\boldsymbol{M}$ are 0. The matrix completion problem can be expressed as a rank minimization problem [17]:

$$\begin{aligned} &\textit{minimize rank}(\boldsymbol{X}) \\ &\text{s.t.}\boldsymbol{X} \in \mathbb{R}^{m\times n}, \boldsymbol{X}_\Omega = \boldsymbol{M}_\Omega \end{aligned} \tag{1}$$

In the Eq. (1), $\boldsymbol{X}$ is the completed matrix. Since the above rank minimization optimization model is a non-convex NP-hard problem, it has no practical significance and is difficult to solve. Therefore, heuristic algorithms are usually used to minimize the kernel norm of the matrix to solve this problem.

$$\begin{aligned} &\underset{\boldsymbol{X}\in\mathbb{R}^{m\times n}}{\textit{minimize}} \| \boldsymbol{X} \|_* \\ &\text{s.t.}\boldsymbol{X}_\Omega = \boldsymbol{M}_\Omega \end{aligned} \tag{2}$$

Where $\| X \|_*$ represents the sum of singular values in $\boldsymbol{X}$, which is also the kernel norm in the matrix. Given a certain number of random numbers (depending on the matrix size and rank) in M_Ω, Eq. (2) usually has a unique minimum $\boldsymbol{X}$ equal to $\boldsymbol{M}$.

Due to the nature of equation constraints, Eq. (2) is susceptible to noise. Therefore, an algorithm for processing noise measurement is proposed in [18]. This algorithm modifies the equation constraint in (2) to:

$$\| \boldsymbol{X}_\Omega - \boldsymbol{M}_\Omega \|_F \leq \delta \tag{3}$$

Where $\| \cdot \|_F$ is the Frobenius norm, parameter $\delta \geq 0$, and is adjusted according to the degree of measurement noise.

2.1. Matrix model based on multi-phase power distribution system

In this section, we establish a matrix model based on the data characteristics of the power system. Let $\mathscr{B}$ represent a set of buses, where bus 1 is a slack bus, and the rest of the buses $|\mathscr{B}| - 1$ are PQ buses, and they are all non-slack buses. Let $\mathscr{L} \subseteq \mathscr{B} \times \mathscr{B}$ represent a collection of branches. The bus admittance matrix$\mathbf{Y} \in \mathbb{C}^{|\mathscr{B}|\times|\mathscr{B}|}$ is described as below:

$$\mathbf{Y} = \begin{bmatrix} \mathbf{Y}_{11} \in \mathbb{C} & \mathbf{Y}_{1L} \in \mathbb{C}^{1\times(|\mathscr{B}|-1)} \\ \mathbf{Y}_{L1} \in \mathbb{C}^{(|\mathscr{B}|-1)\times 1} & \mathbf{Y}_{LL} \in \mathbb{C}^{(|\mathscr{B}|-1)\times(|\mathscr{B}|-1)} \end{bmatrix}$$

Let $\mathrm{v} \in \mathbb{C}^{|\mathscr{B}|}$be the bus voltage and $\mathrm{s} \in \mathbb{C}^{|\mathscr{B}|}$be the injected power. Assuming that the slack bus voltage $\mathbf{V}_1$ and injection power $\mathbf{S}_1$ are known, the non-slack bus voltage $\mathbf{V}$-$_1$ and injection power $\mathbf{S}$-$_1$ are provided by the supervised control and data acquisition (SCADA) system and are partially missing. Let $\mathbf{i}_{ft}$ represent the current in each branch, f represents the beginning of the branch, and t represents the end of the branch.

The configuration of data matrix $\boldsymbol{M}$ and the completed matrix $\boldsymbol{X}$ is adaptable to various measurement scenarios, depending on the specific characteristics of the collected data (e.g., the types and scales of measurements). Here introduces two principal organizational paradigms: (1) a branch-oriented structure and (2) a bus-oriented structure. Importantly, these configurations represent just two possible formulations, the matrix completion framework remains applicable to any measurement arrangement where linear correlations exist among the data points, enabling flexible adaptation to diverse measurement distributions in practical applications.

1) branch-oriented data distribution:

The branch-oriented measurement matrix $\boldsymbol{M}$ is structured with rows representing individual power system branches and columns containing their associated electrical parameters. This configuration enables simultaneous integration of both branch-level measurements and nodal measurement data. Each row vector incorporates the following measurement set for its respective branch $(f, t) \in \mathscr{L}$:

$$\left[\Re(\mathrm{v}_f), \Im(\mathrm{v}_f), |\mathrm{v}_f|, \Re(s_f), \Im(s_f), \Re(\mathrm{v}_t), \Im(\mathrm{v}_t), |\mathrm{v}_t|, \Re(s_t), \Im(s_t), \Re(\mathrm{i}_{ft}), \Im(\mathrm{i}_{ft})\right]$$

Where $\Re(\cdot)$ and $\Im(\cdot)$ are the real part and imaginary part of the complex variable. $\mathbf{V}_f$ is the voltage at the branch end f; $\mathbf{S}_f$ is the power injected at the branch end f; $\mathbf{V}_t$ is the voltage at the branch end t; $\mathbf{S}_t$ is the power injected at the branch end t.

2) bus-oriented data distribution:

The bus- oriented measurement matrix $\boldsymbol{M}$ is constructed with rows corresponding to individual buses and columns containing their associated electrical measurements. Specifically, each row vector comprises the following measurement parameters for its respective bus $\varepsilon \in \mathscr{B}$:

$$[\Re(\mathrm{v}_\varepsilon), \Im(\mathrm{v}_\varepsilon), |\mathrm{v}_\varepsilon|, \Re(s_\varepsilon), \Im(s_\varepsilon)]$$

While the bus-oriented structure utilizes exclusively nodal measurements, it offers significant computational advantages for large-scale system estimation through its compact matrix representation.

The branch- and bus- oriented structures proposed above ensure that each row represents a location (a branch or a bus), and each column represents a measurement type (voltage, power, etc.). This organization reveals two fundamental correlation types:

Spatial correlation (row-wise): This captures the physical relationships between different locations in the power system. Such correlations depend on the inherent properties of the system, such as the relationship between loads at adjacent buses.

Measurement-type correlation (column-wise): This represents the correlation among different types of measurements (e.g., voltage vs. power). This type of correlation is more universal, as power system equations governing these measurements can often be expressed in (approximately) linear forms [19]. Consequently, these data matrices exhibit low-rank properties.

2.2. Linear power flow constraints

The incorporation of linearized power flow constraints as structural regularization effectively characterizes inter-measurement correlations in the matrix completion model. This approach makes the solution process convex and provides crucial system topology information, thereby enabling matrix recovery with fewer measurements. The specific constraints employed include:

1) Repeat constraints: arising from different constructions of matrix $\boldsymbol{M}$, where certain quantities may appear in multiple locations.

In branch-oriented data distribution, if a bus is in multiple branches, the amount measurements associated with a given bus appear in multiple rows. Therefore, let Λ for inclusion a set of all repeated measurement pairs in $\boldsymbol{M}$, such that $\left((\lambda_1, \lambda_2), (\lambda_1', \lambda_2')\right) \in \Lambda \Leftrightarrow \boldsymbol{M}_{\lambda_1,\lambda_2} \equiv \boldsymbol{M}_{\lambda_1',\lambda_2'}$.

$$\boldsymbol{X}_{\lambda_1,\lambda_2} \equiv \boldsymbol{X}_{\lambda_1',\lambda_2'}, \forall\left((\lambda_1, \lambda_2), (\lambda_1', \lambda_2')\right) \in \Lambda \tag{4}$$

2) Ohm's law constraint: When matrix $\boldsymbol{M}$ incorporates both bus and branch quantities (as in branch- oriented formulations), Ohm's Law provides critical physical constraints expressed through:

Algorithm 1
Alternating minimize matrix completion algorithm process.

input:known element sets Ω, data matrix M, linear constraints A,b, parameter μ,ρ.
initialization: $V^{(0)} \in \mathbb{R}^{r\times n}, \alpha \geq 0$, $k=0$.
1: **for** k=1,…, N **do**
2: $U^{(k)} = \mathrm{argmin}_{U\in\mathbb{R}^{m\times r}} \| U \|_F^2 + \mu \| \left(UV^{k-1}\right)_\Omega - M_\Omega \|_F^2 + \rho \| f_1\left(UV^{k-1}\right) - \left(Af_2\left(UV^{k-1}\right)+b\right) \|_2^2$
3: $V^{(k)} = \mathrm{argmin}_{V\in\mathbb{R}^{r\times n}} \| V \|_F^2 + \mu \| \left(U^kV\right)_\Omega - M_\Omega \|_F^2 + \rho \| f_1\left(U^kV\right) - \left(Af_2\left(U^kV\right)+b\right) \|_2^2$
4: **if** $\alpha < 1$
renew k, α, U^N,V^N
else
$k = k+1$ back to step 2.
5: **end for**
6: **returnX** $= U^NV^N$

$$\left(\mathrm{v}_f - \mathrm{v}_t\right)\mathrm{y}_{ft} = \mathrm{i}_{ft}, \forall (f,t) \in \mathscr{L} \tag{5}$$

Where $\mathbf{y}_{ft}$ is the line admittance. However, due to the presence of measurement noise, using exact equality constraints may render the matrix completion algorithm infeasible. Therefore, Ohm's law of noise adaptability is adopted here.

$$\begin{bmatrix} -\xi_{\mathrm{r},ft} \\ -\xi_{\mathrm{c},ft} \end{bmatrix} \leq \begin{bmatrix} \Re\left(\left(\mathrm{v}_f - \mathrm{v}_t\right)y_{ft} - \mathrm{i}_{ft}\right) \\ \Im\left(\left(\mathrm{v}_f - \mathrm{v}_t\right)y_{ft} - \mathrm{i}_{ft}\right) \end{bmatrix} \leq \begin{bmatrix} \xi_{\mathrm{r},ft} \\ \xi_{\mathrm{c},ft} \end{bmatrix} \tag{6}$$

Where $\xi_{\mathrm{r},ft}$ represents error tolerance of the real part of Ohm's Law, and $\xi_{\mathrm{c},ft}$ represents error tolerance of the complex part of Ohm's Law on the branch $(f,t) \in \mathscr{L}$.

3) Linearized power flow constraints: given the inherent nonlinearity of AC power flow equations, we employ cartesian coordinate linearization to derive tractable constraints. For non-slack bus voltages and power injections, the fllowing approximations are implemented:

$$\mathrm{v}_{-1} \approx B\begin{bmatrix} \Re(s_{-1}) \\ \Im(s_{-1}) \end{bmatrix} + w \tag{7}$$

$$|\mathrm{v}_{-1}| \approx C\begin{bmatrix} \Re(s_{-1}) \\ \Im(s_{-1}) \end{bmatrix} + |w| \tag{8}$$

Let $w = -Y_{LL}^{-1}Y_{L1}\mathrm{v}_1 \in \mathbb{C}^{|\mathscr{B}|-1}$ be the phasor of the no-load voltage of the non-slack bus **V-**$_1$, and define$B, C \in \mathbb{C}^{(|\mathscr{B}|-1)\times 2(|\mathscr{B}|-1)}$.

$$B = \left[Y_{LL}^{-1}diag(\overline{\mathrm{v}_{-1}})^{-1} - jY_{LL}^{-1}diag(\overline{\mathrm{v}_{-1}})^{-1}\right] \tag{9}$$

$$C = diag(|\mathrm{v}_{-1}|)^{-1}\Re(diag(|\overline{\mathrm{v}_{-1}}|)B) \tag{10}$$

Other methods can also be used to obtain linear approximations (9), such as data-driven regression methods [20].

To establish the relationship between the voltages and the power injections at the slack bus, accurate power flow equation is employed:

$$s_1 = \mathrm{v}_1(\overline{Y_{11}}\overline{\mathrm{v}_1} + \overline{Y_{1L}}\overline{\mathrm{v}_{-1}}) \tag{11}$$

Since $\mathbf{V}_1$ is known, this equation is linear with the voltage.

Then change the above equations to noise adaptability constraint:

$$\begin{bmatrix} -\tau_{\mathrm{r}} \\ -\tau_{\mathrm{c}} \end{bmatrix} \leq \begin{bmatrix} \Re\left(\mathrm{v}_{-1} - \left(B\begin{bmatrix} \Re(s_{-1}) \\ \Im(s_{-1}) \end{bmatrix} + w\right)\right) \\ \Im\left(\mathrm{v}_{-1} - \left(B\begin{bmatrix} \Re(s_{-1}) \\ \Im(s_{-1}) \end{bmatrix} + w\right)\right) \end{bmatrix} \leq \begin{bmatrix} \tau_{\mathrm{r}} \\ \tau_{\mathrm{c}} \end{bmatrix} \tag{12}$$

$$-\gamma \leq |\mathrm{v}_{-1}| - \left(C\begin{bmatrix} \Re(s_{-1}) \\ \Im(s_{-1}) \end{bmatrix} + |w|\right) \leq \gamma \tag{13}$$

$$\begin{bmatrix} -\alpha_{\mathrm{r}} \\ -\alpha_{\mathrm{c}} \end{bmatrix} \leq \begin{bmatrix} \Re(s_1 - \mathrm{v}_1(\overline{\mathrm{Y}}_{11}\overline{\mathrm{v}}_1 + \overline{\mathrm{Y}}_{1L}\overline{\mathrm{v}}_{-1})) \\ \Im(s_1 - \mathrm{v}_1(\overline{\mathrm{Y}}_{11}\overline{\mathrm{v}}_1 + \overline{\mathrm{Y}}_{1L}\overline{\mathrm{v}}_{-1})) \end{bmatrix} \leq \begin{bmatrix} \alpha_{\mathrm{r}} \\ \alpha_{\mathrm{c}} \end{bmatrix} \tag{14}$$

Where $\tau_{\mathbf{r}},\tau_{\mathbf{c}},\gamma,\alpha_{\mathbf{r}},\alpha_{\mathbf{c}}$are error tolerances.

Let $s_{-1}^{(T)}$, $\mathrm{v}_{-1}^{(T)}$, $w^{(T)}$, $h^{(T)}$, $B^{(T)}$, $C^{(T)}$ represent **S-**$_1$, **V-**$_1$, w, h, B, C corresponding to time T. To simplify subsequent expression, the following model is used to express the above linear model:

$$\boldsymbol{y} \approx \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b} \tag{15}$$

Where $y = \begin{bmatrix} \mathrm{v}_{-1}^{(T)} \\ |\mathrm{v}_{-1}^{(T)}| \end{bmatrix}$; $A = \left[\left(B^{(T)}\right)^{\mathrm{T}}\left(C^{(T)}\right)^{\mathrm{T}}\right]^{\mathrm{T}}$; $x = \left[h^{(T)}\right]$ $(\mathfrak{h} = [\Re(s_{-1})\Im(s_{-1})]^{\mathrm{T}})$; $b = \begin{bmatrix} \boldsymbol{w}^{(T)} \\ |\boldsymbol{w}^{(T)}| \end{bmatrix}$.

The complete matrix completion model with power flow constraints is expressed as:

$$\min \| \boldsymbol{X} \|_* + \frac{\mu}{2}\| \boldsymbol{X}_\Omega - \boldsymbol{M}_\Omega \|_F^2 + \frac{\rho}{2}\| y - (Ax+b) \|_2^2 \tag{16}$$

Where $\boldsymbol{X} \in \mathbb{R}^{m\times n}$, $y \in \mathbb{C}^{\frac{2}{5}m\times n}$, $x \in \mathbb{R}^{\frac{2}{5}m\times n}$, s.t. $y = \left[z_1^{\mathrm{I}}X \;\; z_2^{\mathrm{I}}X \;\; \cdots \;\; z_{3T}^{\mathrm{I}}X\right]^{\mathrm{T}}$,$x = \left[c_1^{\mathrm{I}}X \;\; c_2^{\mathrm{I}}X \;\; \cdots \;\; c_{2T}^{\mathrm{I}}X\right]^{\mathrm{T}}$. $\mu,\rho>0$ are penalty parameters; $z_1, z_2, \cdots, z_{3T} \in \mathbb{C}^m$ and $c_1, c_2, \cdots, c_{2T} \in \mathbb{R}^m$ are constants, and their values can be described through the following specific forms of X, y, and x:

$$\boldsymbol{X} = \begin{bmatrix} \Re\left(\boldsymbol{v}_{-1}^{(T)}\right) \\ \Im\left(\boldsymbol{v}_{-1}^{(T)}\right) \\ \left|\boldsymbol{v}_{-1}^{(T)}\right| \\ \Re\left(\boldsymbol{s}_{-1}^{(T)}\right) \\ \Im\left(\boldsymbol{s}_{-1}^{(T)}\right) \end{bmatrix}^{\mathrm{T}}, y = \begin{bmatrix} \Re\left(v^{(T)}\right) + j\Im\left(v^{(T)}\right) \\ \left|v^{(T)}\right| \end{bmatrix}, x = \begin{bmatrix} \Re\left(s_{-1}^{(T)}\right) \\ \Im\left(s_{-1}^{(T)}\right) \end{bmatrix}$$

2.3. Model solution

The alternating minimization algorithm, which employs low-rank matrix fitting via LMaFit [21] to update variables in an alternating manner, stands as one of the most efficient and accurate methods for addressing matrix completion problems [22], and is capable of rapidly addressing large-scale matrix completion tasks. The model-solving procedure is described below.

1) First, any matrix $\boldsymbol{X} \in \mathbb{R}^{m\times n}$ with a rank $\mathbf{r}$ can be represented as the matrix product form of $U \in \mathbb{R}^{m\times r}$ and $V \in \mathbb{R}^{r\times n}$, that is, its core norm can be represented as the following using the Frobenius norm of $U \in \mathbb{R}^{r\times n}$ and $V \in \mathbb{R}^{r\times n}$ as follow:

$$\| \boldsymbol{X} \|_* := \mathrm{argmin}\frac{1}{2}\left(\| U \|_F^2 + \| V \|_F^2\right) \tag{17}$$

For extremely small rank $\mathbf{r}$ ($\mathrm{r} \ll \min\{m,n\}$), using the above characterization of the kernel norm can significantly reduce the size of the problem. Then re-express the matrix completion model as follows:

$$\begin{aligned} &\mathrm{minimize}\frac{1}{2}\left(\| U \|_F^2 + \| V \|_F^2\right) + \frac{\mu}{2}\| \boldsymbol{X}_\Omega - \boldsymbol{M}_\Omega \|_F^2 + \frac{\rho}{2}\| f_1(UV) - (Af_2(UV)+b) \|_2^2 \\ &\mathrm{s.t.} f_1(UV) = [z_1^{\mathrm{I}}UV \; z_2^{\mathrm{I}}UV \; \cdots \; z_{2T-1}^{\mathrm{I}}UV \; z_{2T}^{\mathrm{I}}UV]^{\mathrm{T}} \\ &f_2(UV) = [c_1^{\mathrm{I}}UV \; c_2^{\mathrm{I}}UV \; \cdots \; c_{2T-1}^{\mathrm{I}}UV \; c_{2T}^{\mathrm{I}}UV]^{\mathrm{T}} \end{aligned} \tag{18}$$

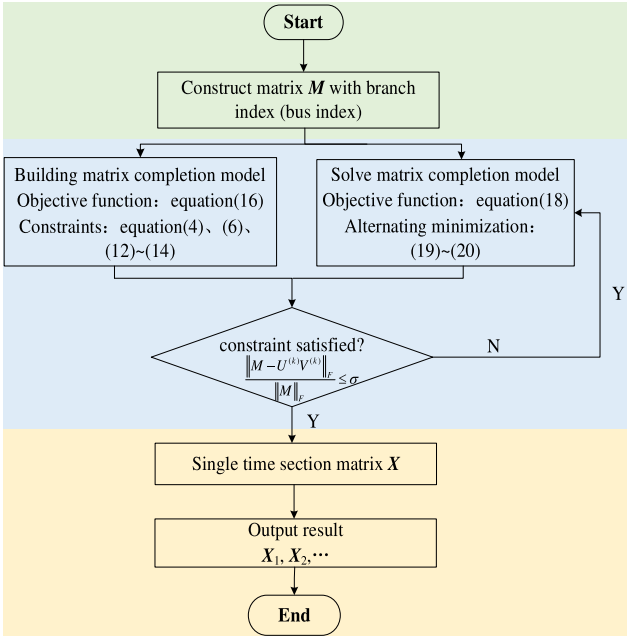


Fig. 1. Flow chart of matrix completion model.

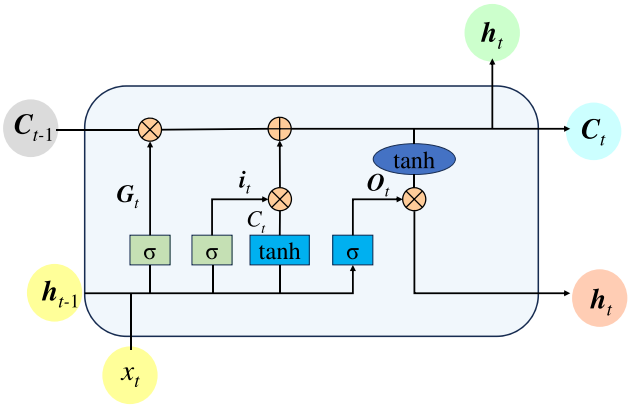


Fig. 2. LSTM network structure.

2) Next, the alternating minimization algorithm sets the number of iterations $k = 1, \ldots, N$ in the case of fixed matrix rank **r**, and alternately solves U and V:

$$U^{(k)} = \underset{U \in \mathbb{R}^{m \times r}}{\operatorname{argmin}} \| U \|_F^2 + \mu \| \left(UV^{k-1}\right)_\Omega - M_\Omega \|_F^2 + \rho \| f_1\left(UV^{k-1}\right) - \left(Af_2\left(UV^{k-1}\right) + b\right) \|_2^2 \quad (19)$$

$$V^{(k)} = \underset{V \in \mathbb{R}^{r \times n}}{\operatorname{argmin}} \| V \|_F^2 + \mu \| \left(U^k V\right)_\Omega - M_\Omega \|_F^2 + \rho \| f_1\left(U^k V\right) - \left(Af_2\left(U^k V\right) + b\right) \|_2^2 \quad (20)$$

3) Finally, when the iteration result satisfies $\frac{\|\boldsymbol{M} - U^{(k)}V^{(k)}\|_F}{\|\boldsymbol{M}\|_F} \leq \alpha$, iteration is stopped. The residual threshold is $\alpha \in [0, 1)$ and can be adjusted depending on the data situation [21]. Solution to the time T complete matrix is $\mathbf{X} = U^N V^N$.

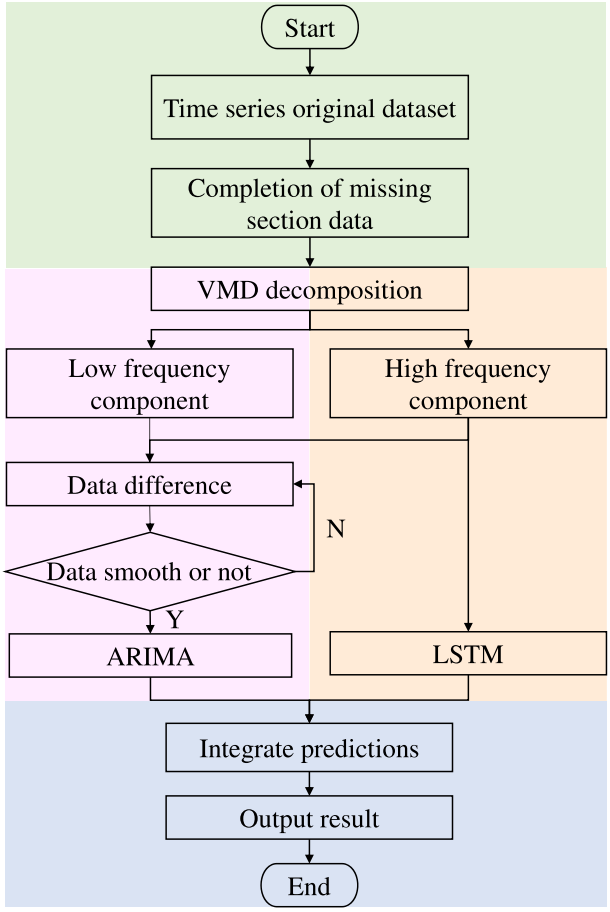


Fig. 3. Flow chart of MC-VMD-ARIMA-LSTM prediction model,.

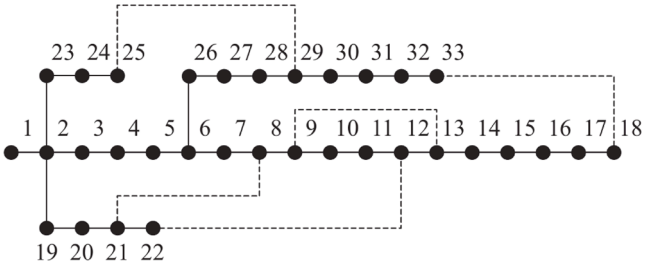


Fig. 4. Topology of IEEE 33-bus distribution network,.

The following is the pseudocode for solving the algorithm Algorithm 1.

3. Multi-timescale data completion model

The digital transformation of power systems has driven widespread deployment of advanced measurement devices to enhance grid observability [23]. Current data acquisition systems typically incorporate: 1) supervised control and data acquisition system (SCADA), and the sampling intervals usually range from seconds to minute levels. 2) Smart meter or advanced metrology infrastructure (AMI) data, the sampling interval is usually about 15 minutes. 3) The voltage phasor measurement value collected by the phasor measurement unit (PMU) is usually

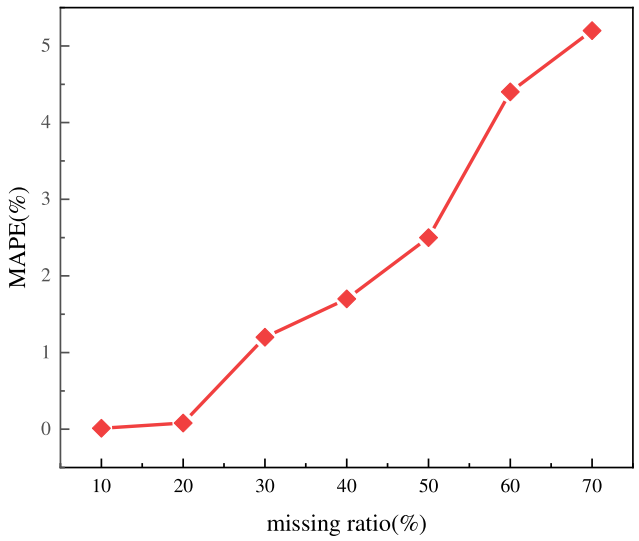


Fig. 5. Completion result of power flow data on IEEE 33-bus distribution network,.

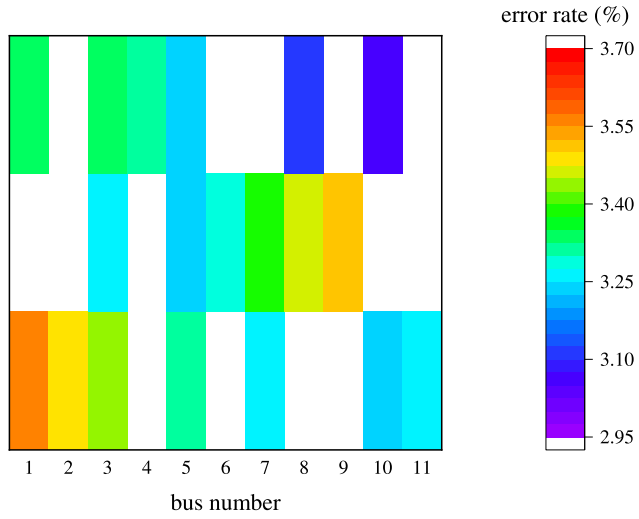


Fig. 6. Thermal diagram of the voltage amplitude completion error rate of IEEE 33-bus distribution network,.

sampled at a rate of 120 samples per second.

The existing measurement infrastructure exhibits several technical limitations that impact data quality and system observability. Sampling rates remain insufficient for capturing dynamic system conditions, while inconsistent time synchronization between measurement devices introduces phase measurement errors. Furthermore, extended periods of missing data segments frequently occur due to communication interruptions or device failures. These data quality issues necessitate advanced algorithmic approaches capable of processing and correlating multi-time scale measurement to maintain accurate state estimation under realistic operating conditions.

3.1. Integration of matrix completion with ARIMA (MC-ARIMA)

The moving average (MA) technique serves as a fundamental analytical tool in multi-timescale temporal analysis. Operationally

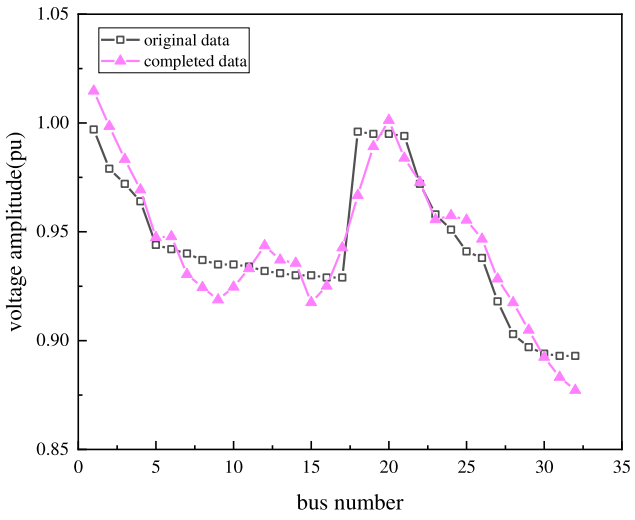


Fig. 7. Completion result of the voltage amplitude at a certain time in IEEE 33-bus distribution network,.

grounded in the equally-weighted averaging principle, this method implements a sliding window mechanism that systematically incorporates incoming observations while discarding outdated data points through sequential updates. This helps eliminate random fluctuation factors, identify the trend of the matter, and make predictions accordingly [24].

The Autoregressive Integrated Moving Average (ARIMA) methodology extends conventional MA techniques by incorporating differencing operations and autoregressive components. This approach systematically transforms non-stationary time series into stationary sequences through sequential differencing, enabling effective analysis of multi-time scale data in distribution networks. In distribution network applications, data may fluctuate short-term due to measurement errors or external interference. ARIMA handles random noise with moving average (MA) while extracting the autocorrelation between past and current observations through Autoregressive (AR).

$$\boldsymbol{X}_{T+1} = \varphi_1 \boldsymbol{X}_T + \varphi_2 \boldsymbol{X}_{T-1} + \cdots + \varphi_p \boldsymbol{X}_{T+1-p} + \eta_{T+1} - \theta_1 \eta_T - \cdots - \theta_q \eta_{T+1-q} \tag{21}$$

Where $\boldsymbol{p}$ and $\boldsymbol{q}$ are autoregressive orders and moving average orders, $\boldsymbol{\eta}$ is noise sequence, $\varphi_1\varphi_2\cdots\varphi_p$ is autoregressive coefficient, and $\theta_1\theta_2\cdots\theta_q$ is moving average coefficient.

When historical data exhibits relatively stable changes over a period of time, lower orders are typically sufficient. Conversely, when fluctuations are intense, the order should be appropriately increased to prevent overfitting to noise, which could lead to a decrease in the accuracy of the results.

3.2. Integration of matrix completion with LSTM (MC-LSTM)

The ARIMA methodology demonstrates computational efficiency advantages for real-time/near-real-time data completion tasks in distribution networks, owing to its polynomial time complexity (O(n)) and minimal memory footprint. When applied to modern distribution networks with high penetration of DERs, ARIMA may be inadequate in computational efficiency.

LSTM (Long Short-Term Memory) is a deep learning algorithm based on the RNN (Recurrent Neural Network) structure. It addresses the issues of gradient explosion and vanishing gradients over time by introducing gating mechanisms and memory units, making it effective for

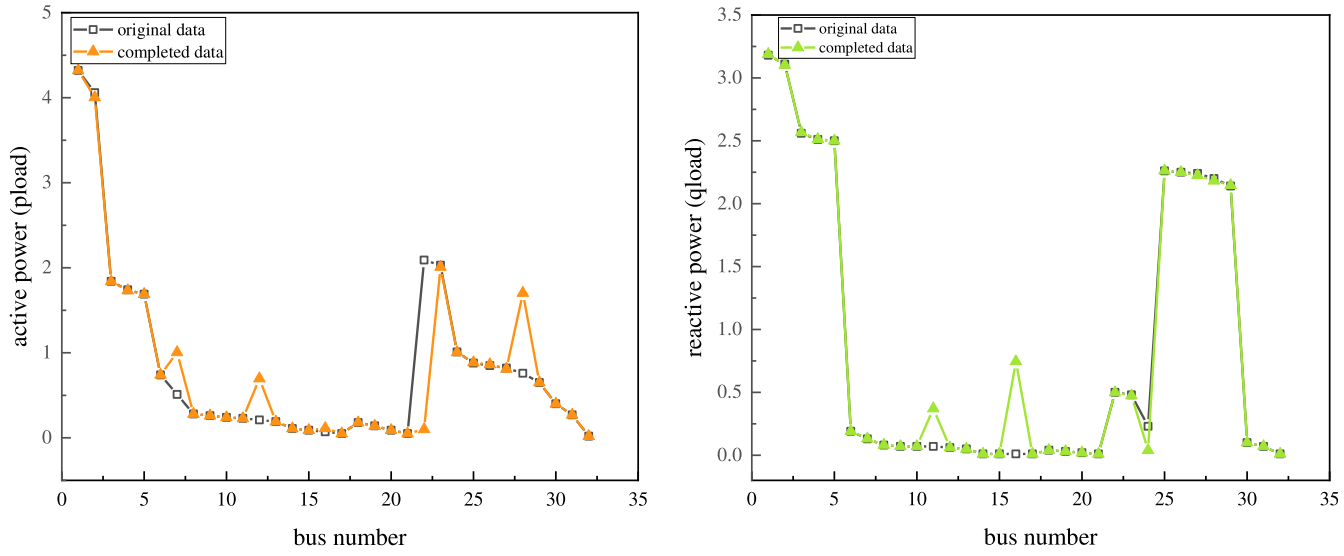


Fig. 8. Completion result of a heavy-load time in the IEEE 33-bus distribution network.

Table 1
Key references analysis.

Reference	Major Findings	Relevance to Our Work
Refs [2,5–9]	Low-observability of distribution networks	Motivates problem significance
Refs [3,4]	WLS was proposed	Initial method for state estimation
Refs [10,13, 14]	Use neural networks for state estimation	Solve the problem of large-scale estimation
Refs [11,15]	Three-phase distribution network state estimation	Consider the linear correlation of the data
Refs [12,16]	Optimization under poor data quality	Support poor data quality state estimation

Table 2
Temporal power flow data completion results.

Time step	Missing ratio		MAPE (%)	MAE($\times 10^{-2}$)
1	30 %-40 %		0.8	1.15
2			0.9	1.18
3			1.2	1.32
4			1.2	1.32
5			0.9	1.17
6			1.1	1.19
7	100 %	MC	—	
		MC-ARIMA	1.4	2.37

Table 3
Running time of IEEE 33-bus distribution network data completion algorithm.

Missing ratio (%)	10	20	30	40	50	60	70
running time (s)	2.8	3.3	4.8	5.6	7.2	7.6	8.5

Table 4
Origin input data statistics.

Types of statistical data	Maximum value	Minimum value	Average
A-phase current(A)	12.35	34.82	25.09
Active power(kW)	136.60	31.41	65.77

Table 5
Center frequencies of each IMF at different $\boldsymbol{K}$ values.

K	IMF 1	IMF 2	IMF 3	IMF 4	IMF 5
2	1.46×10^{-4}	0.2424			
3	1.46×10^{-4}	0.1563	0.3257		
4	1.46×10^{-4}	0.1141	0.2445	0.3720	
5	1.46×10^{-4}	0.0908	0.1931	0.2941	0.4501
6	1.46×10^{-4}	0.0751	0.1607	0.2407	0.3298

handling long sequence data. The LSTM model incorporates input gates, forget gates, and output gates within its neurons, enabling selective filtering of historical information [25]. These gates are controlled by the Sigmoid function and combined with the tanh function. The mathematical formulation of the LSTM model is as follows:

1) Forget Gate:

$$G_{T+1} = \sigma(W_F \cdot [h_T, x_{T+1}] + b_F) \tag{22}$$

Where $\boldsymbol{G_{T+1}}$ is the state matrix of the forget gate at time step $T+1$; $\boldsymbol{h_T}$ is the state matrix of the forget gate at the previous moment time step T; $\boldsymbol{x_{T+1}}$ is the input value at time step $T+1$; $\sigma(\cdot)$is the activation function; $\boldsymbol{W_F}$ is the weight matrix of the forget gate; $\boldsymbol{b_F}$ is the constant parameter matrix of the forget gate.

2) Input Gate:

$$i_{T+1} = \sigma(W_s \cdot [h_T, x_{T+1}] + b_s) \tag{23}$$

Where $\boldsymbol{i_{T+1}}$ is the state matrices of the input gate at time step $T+1$; $\boldsymbol{W_S}$ is the weight matrices for the input gate; $\boldsymbol{b_s}$ is the constant parameter matrices.

3) Candidate Cell State:

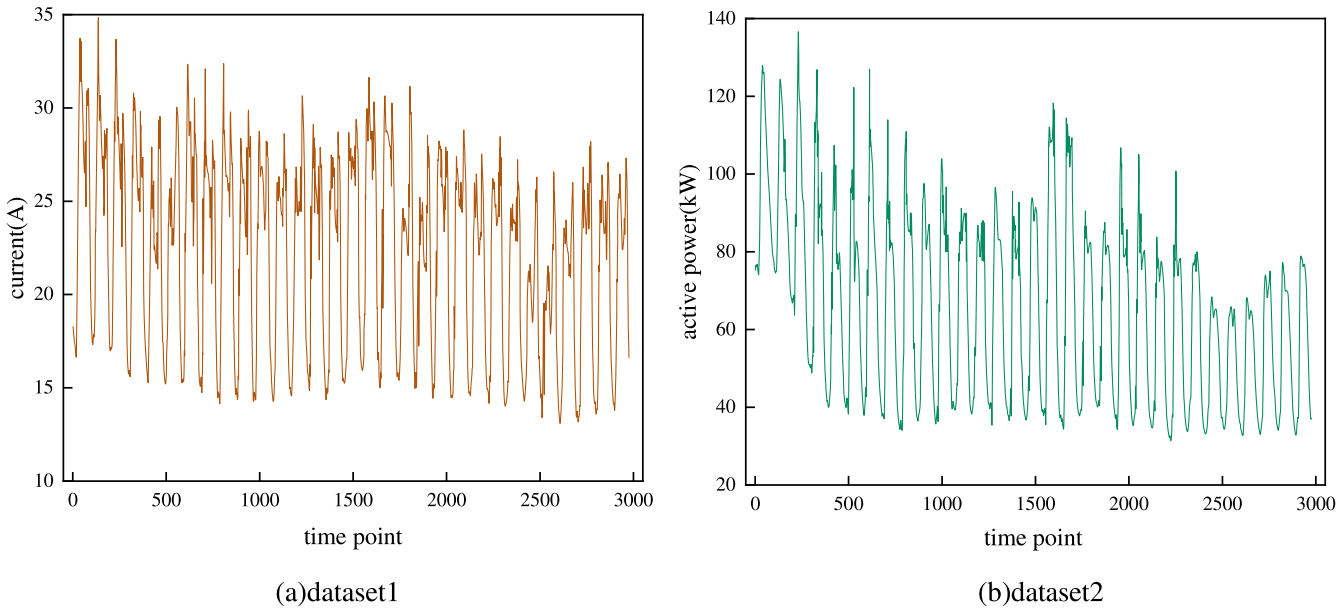


Fig. 9. Origin input dat .

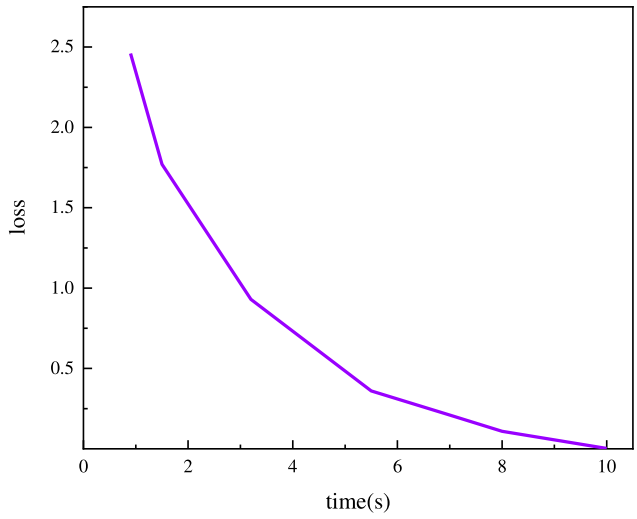


Fig. 10. Iterative convergence curve.

$$\widetilde{C_{T+1}} = \tanh(W_c \cdot [h_T, x_{T+1}] + b_c) \tag{24}$$

Where tanh is hyperbolic tangent activation function; $\widetilde{C_{T+1}}$is the state matrices of the intermediate input at time step T+1; $\boldsymbol{b_c}$ is the weight matrices for the intermediate output.

4) Cell State Update:

$$C_{T+1} = G_{T+1} C_T + i_{T+1} \widetilde{C_{T+1}} \tag{25}$$

Updates the cell state by combining the previous cell state and the candidate state.

5) Output Gate:

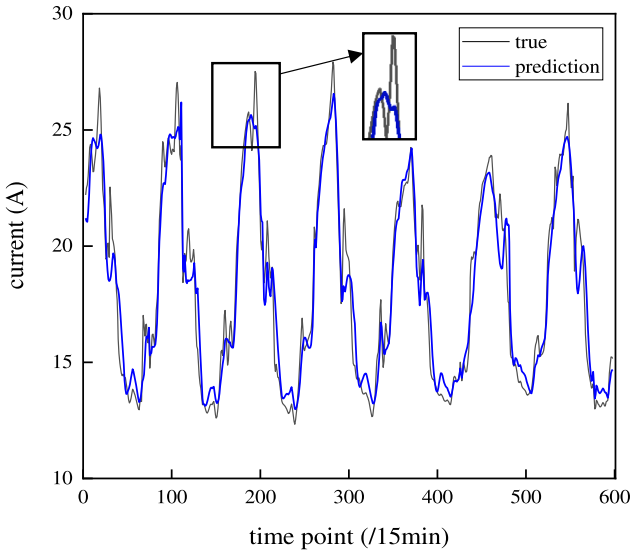


Fig. 11. MC-LSTM prediction result of dataset 1.

$$O_{T+1} = \sigma[W_o[h_T, x_{T+1}] + b_o] \tag{26}$$

where $\boldsymbol{O_{T+1}}$ is the state matrix of the output gate at time step T+1; $\boldsymbol{W_O}$ is the weight matrix of the output gate, and $\boldsymbol{b_o}$ is the constant parameter matrix of the output gate.

6) Hidden State:

$$H_{T+1} = O_{T+1} \tanh(C_{T+1}) \tag{27}$$

3.3. Integration of matrix completion with VMD-ARIMA-LSTM (MC-VMD-ARIMA-LSTM)

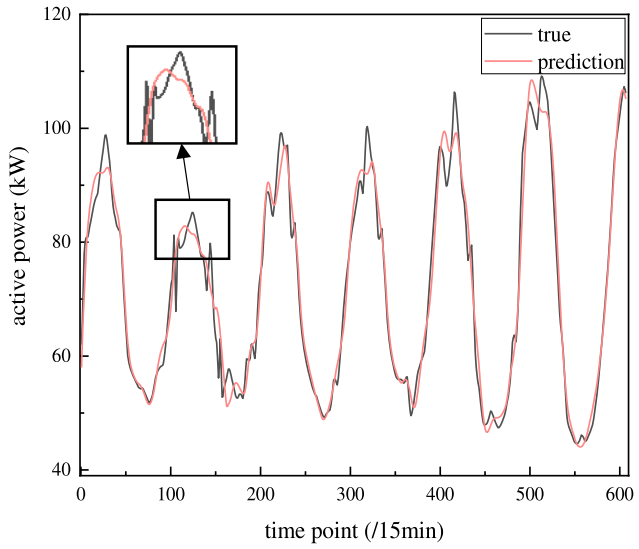


Fig. 12. MC-LSTM prediction result of dataset 2.

The inherent stochasticity and multidimensional complexity of distribution network measurement data contribute significantly to pervasive data incompleteness in modern power systems. Conventional single-model prediction approaches prove inadequate for capturing the spatiotemporal evolution patterns influenced by dynamic load variations and renewable generation intermittency. This often results in poor prediction performance or even the inability of the prediction algorithm to operate.

Variational mode decomposition (VMD) is a novel adaptive and non-recursive time-domain processing algorithm proposed by Dragomiretskiy K et al [26]. This non-recursive algorithm operates through the decomposition of raw time-series data into multiple intrinsic mode functions (IMFs) with distinct central frequencies, effectively addressing the complexity and nonlinearity inherent in distribution network measurements. While ARIMA provides computational efficiency for linear trend modeling, its inability to capture nonlinear dynamics limits its standalone utility. LSTM networks, as a subclass of recurrent neural networks, demonstrate superior performance in modeling temporal nonlinearities through their gated memory mechanisms. The combination of ARIMA and LSTM combines the advantages of linear and nonlinear models and improves the accuracy of model prediction [27]. Therefore, this paper presents an integrated forecasting model MC-VMD-ARIMA-LSTM.

VMD decomposes the original signal into $\boldsymbol{K}$ modal components, each of which is redefined as a modal function $\boldsymbol{u_{K(T)}}$ of the input signal, calculated as follows:

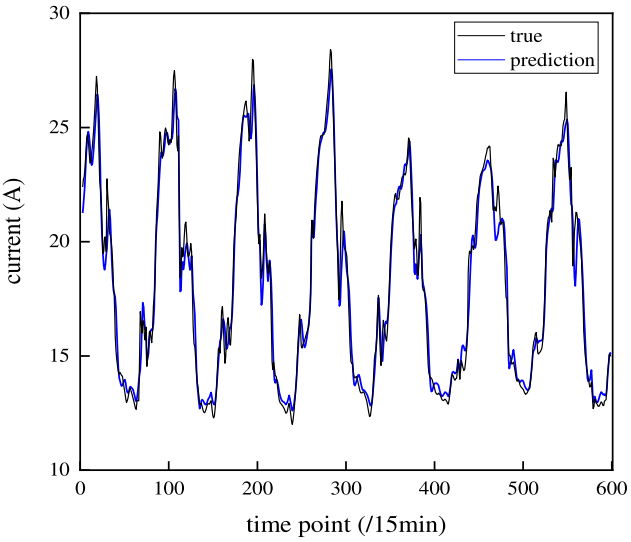


Fig. 14. MC-VMD-ARIMA-LSTM prediction result of dataset 1.

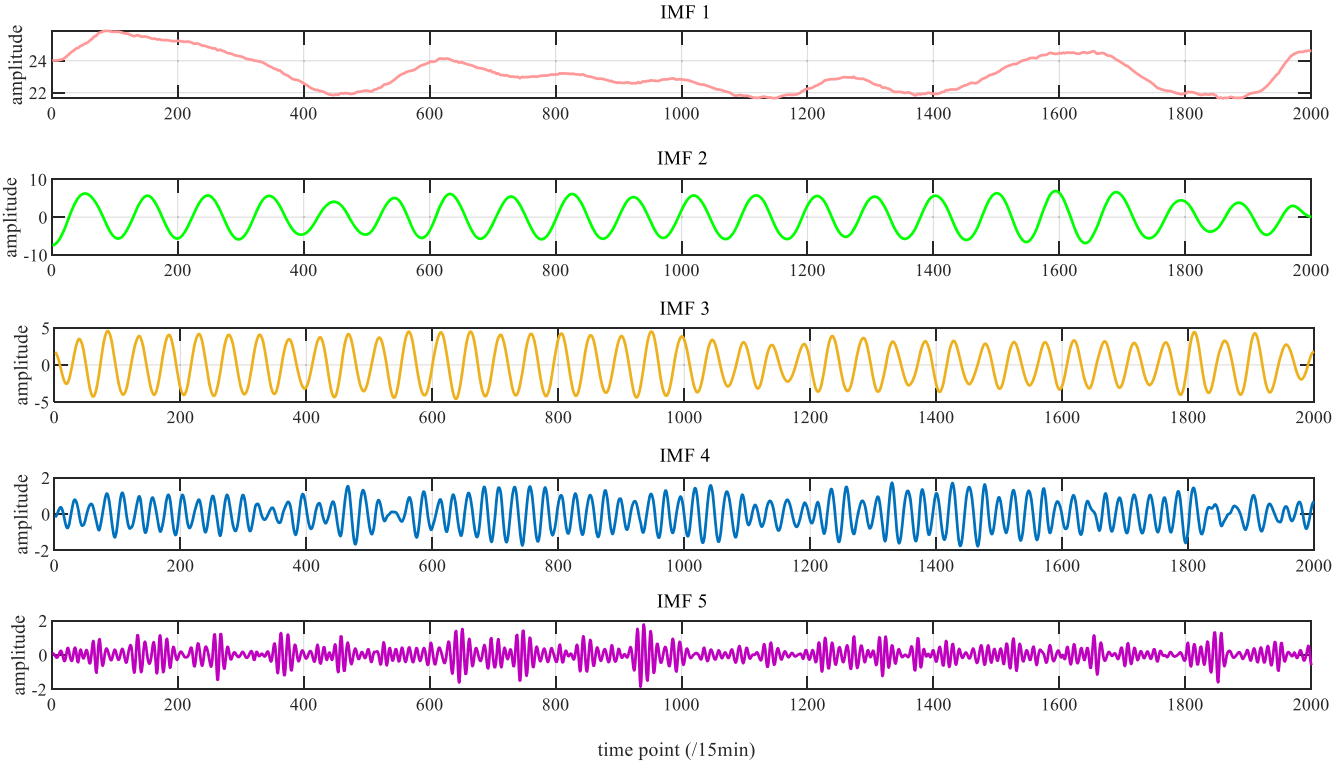


Fig. 13. VMD decomposition result diagram.

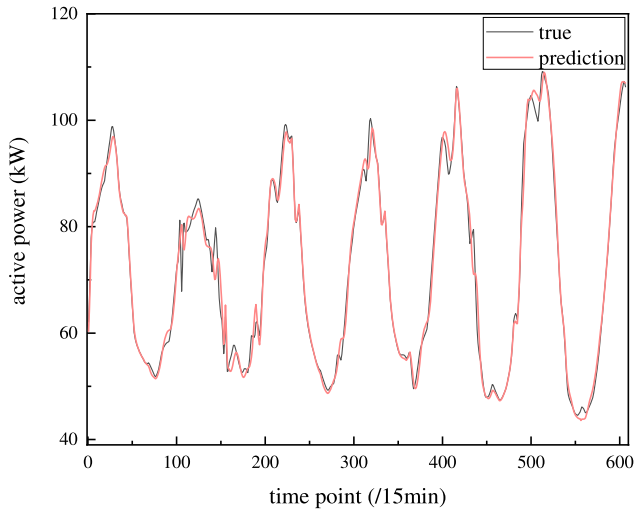


Fig. 15. MC-VMD-ARIMA-LSTM prediction result of dataset 2.

$$u_K(T) = A_K(T)\cos(\phi_K(T)) \tag{28}$$

Where $\boldsymbol{A_{K(T)}}$ represents the amplitude; $\boldsymbol{\phi_{K(T)}}$ represents the phase.

The corresponding bandwidth-constrained variational problem is formulated as follows:

$$\begin{cases} \min\limits_{v_K u_K}\left\{\sum\limits_{K'=1}^{K} \| \partial_T\left[\left(\delta_K(T)+\dfrac{j}{\pi T}\right)u_K(T)\right]e^{-2\pi j\omega T}\|_2^2\right\} \\ \sum\limits_{K'=1}^{K} u_K(T) = f(T) \end{cases} \tag{29}$$

Where $\boldsymbol{f(T)}$ represents the original signal to be decomposed; $\boldsymbol{\delta_K(T)}$ represents the Dirac function; $\boldsymbol{\omega}$ is the central frequency; $\boldsymbol{K}$ is the number of modes.

3.4. Model solution

The *MC-VMD-ARIMA-LSTM* prediction process is as follows:

Step 1: Complete the original time series data section through the matrix completion algorithm;
Step 2: Denoising and decomposing the completed time series data through VMD algorithm to obtain a total of $\boldsymbol{K}$ low-frequency and high-frequency components;
Step 3: Use differential processing of noise in low-frequency and high-frequency components, and input the smoothed components into ARIMA and LSTM prediction models;
Step 4: Integrate $\boldsymbol{K}$ component data to obtain the final predicted result.

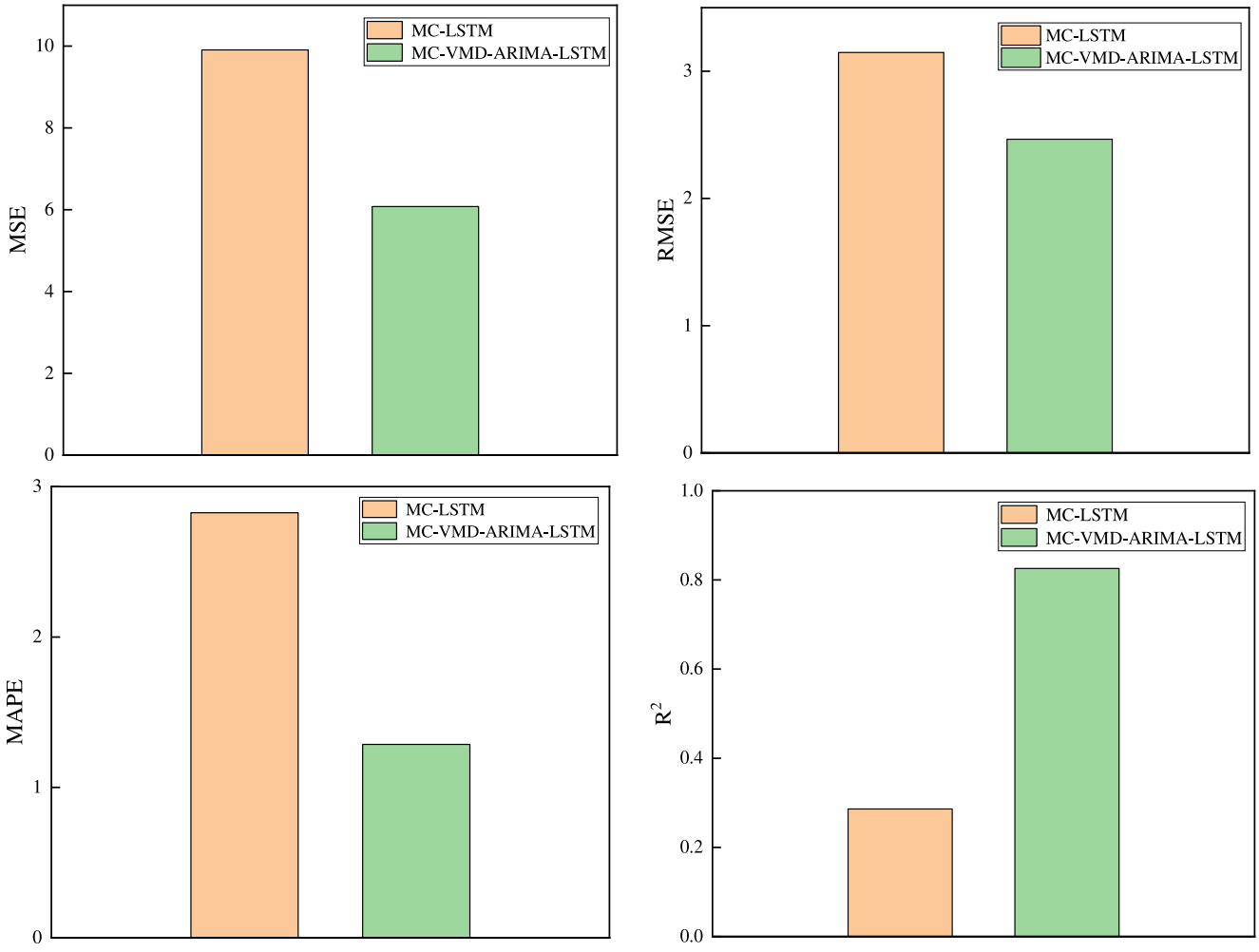


Fig. 16. Error comparison diagram of MC-LSTM and MC-VMD-ARIMA-LSTM.

3.5. Model evaluation

To visually assess the prediction accuracy of the model, the Root Mean Square Error (RMSE), Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE) and R-Square (R^2) are selected as evaluation metrics for the predictive model. Metrics closer to zero indicate higher prediction accuracy, with the formulas as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{m \times n} \sum_{q=1}^{m \times n} \left(x_q - \widehat{x}_q\right)^2} \tag{30}$$

$$\mathrm{MSE} = \frac{1}{m \times n} \sum_{q=1}^{m \times n} \left(x_q - \widehat{x}_q\right)^2 \tag{31}$$

$$\mathrm{MAPE} = \frac{1}{m \times n} \sum_{q=1}^{m \times n} \left| \frac{x_q - \widehat{x}_q}{x_q} \right| \times 100\% \tag{32}$$

$$\mathrm{R}^2 = 1 - \frac{\sum_{q=1}^{m \times n} \left(x_q - \widehat{x}_q\right)^2}{\sum_{q=1}^{m \times n} \left(x_q - \overline{x}_q\right)^2} \tag{33}$$

Where $m \times n$ is the number of data points, $x_q$represents the actual value of the qth data point and$\widehat{x}_q$ represents the estimated value of the qth data point.

4. Experimental results and analysis

This section presents a comprehensive performance evaluation of three proposed methodologies: MC-ARIMA, MC-LSTM and MC-VMD-ARIMA-LSTM. We run the following training process on our own PC. The configuration of our CPU is Intel(R) Core(TM) Ultra 9 285H 2.90 GHz with 32GB RAM. The implementation of the proposed models is based on PyCharm 2024.1.

4.1. Case Setting

Case 1: MC-ARIMA

Case 1 uses the standard IEEE 33-bus distribution network demonstrate the performance of matrix completion model. IEEE 33-bus distribution network has 32 lines and 5 tie-line branches (25-29, 8-21, 12-22, 9-15, and 18-33), with a base voltage of 12.66 kV. Related parameters can be found in [28].

The power flow data categories on the 33-bus distribution network comprise several key electrical parameters, including the voltage amplitude, voltage phase angle, active power and reactive power injected, and active power and reactive power output by the buses Figs. 1-4.

1) Random power flow data missingness of single time section

The performance evaluation of our proposed alternating minimization matrix completion algorithm on the IEEE 33-bus distribution network is presented in Fig. 5. To rigorously test the algorithm's robustness, we artificially removed 10 % to 70 % of the power flow dataset and applied the matrix completion algorithm for data matrix reconstruction. When the number of known data is more than 70 %, the algorithm achieves exceptional recovery accuracy with the MAPE of the data completion result below 1 %; in scenarios with below 70 % observable data, the method maintains reliable performance, constraining MAPE within 10 % across all test cases. Further validation is provided in Figs. 6 and 7, which compare the completion results of the voltage amplitudes against standard values under 30 %-40 % data missing conditions.

Fig. 8 shows the comparison of the completion results of the injection of active and reactive power at certain heavy load moment in the IEEE 33-bus distribution network with the standard data of the case. It can be seen that when more than half of the power flow data are missing, the algorithm proposed in this paper can still generate accurate completed data. Notably, the methodology achieves precise data matrix reconstruction (MAPE <5 % for active power and reactive power) even during sustained heavy loading periods (bus loading factors >85 %), confirming its practical viability for various network conditions.

2) Temporal power flow data missingness

The alternating minimization matrix completion algorithm encounters inherent limitations when processing temporal data in power flow measurements. To address this challenge, we develop MC-ARIMA model. This integrated approach systematically resolves continuous temporal data missingness through spatiotemporal correlation exploitation.

The results from the table above indicate that when the proportion of missing data exceeds 30 %. When the missing data proportion surpasses 30 %, the algorithm achieves a mean absolute percentage error (MAPE) of approximately 1 % in data completion. Under the condition of complete data absence at the next time point, it can be integrated with the ARIMA algorithm to predict all types of power flow data at that moment, achieving a relative accuracy of about 1.4 %, which demonstrates a high level of reliability.

The table above presents the average computation time achieved by the completion algorithm under different missing ratios (each ratio condition was tested over 30 runs). The processing can be efficiently handled by a modern single-core CPU without significant bottlenecks.

Case 2:

- MC-LSTM

While the ARIMA methodology demonstrates efficacy in short-term temporal data imputation, it faces inherent limitations when addressing extended-period forecasting challenges characterized by prolonged data missingness and high-dimensional temporal datasets. These constraints stem fundamentally from ARIMA's modeling assumptions and finite memory window, which prove inadequate for capturing long-range temporal dependencies in distribution network measurements. Therefore, LSTM algorithm is taken into consideration, which enables robust handling of multi-scale temporal correlations spanning hours to days.

Case 2 selected certain regional 10 kV distribution systems (region 1) as experimental subjects to verify the performance of the MC-LSTM model. The sample datasets (dataset 1 and dataset 2) consist of the A-phase current measurement data (dataset 1) and active power measurement data (dataset 2) collected from the load switches at the feeder outlets of region 1, respectively, from March 1st to March 31th, 2022. The sampling interval is set at 15 minutes, resulting in 96 sample points collected each day. Consequently, each dataset comprises a total of 2976 measurement data points.

In case 2, the dataset was configured in an 8:2 training-test split ratio with 40 %-50 % of the training data randomly removed to simulate realistic measurement loss condition. The Fig. below illustrates the iterative convergence curve of the MC-LSTM algorithm when processing the dataset 1. Considering the characteristics of power flow data, this study employs torch.optim.Adam as the optimizer. It automatically adapts learning rates for each input variable of the objective function and computes update steps by comprehensively integrating first-moment estimates (mean of gradients) and second-moment estimates (uncentered variance of gradients). The computational efficiency metrics reveal complete convergence in approximately 10 seconds, with

total execution time consistently maintained under 20 seconds across multiple trials. This efficiency persists under varying initialization conditions.

The following Fig. shows the processing results of the MC-LSTM algorithm on Dataset 1 and Dataset 2. As illustrated in the Fig., the predicted values generally align with the trend of the actual measurements. However, there are some discrepancies in the details, such as instances where the values change abruptly over a short period. Additionally, due to the loss of data across multiple consecutive time points, the MC algorithm's completion will inevitably introduce some noise, which subsequently causes the LSTM algorithm to produce biased predictions.

Tables 1-4

- MC-VMD-ARIMA-LSTM

To further enhance the prediction accuracy, VMD is incorporated, which is suitable for nonlinear time series signal processing, after the data imputation. Since VMD can flexibly adjust decomposition parameters, the decomposition results are less affected by signal length and sampling rate. Subsequently, this research combined ARIMA, which is adept at short-term forecasting, with LSTM, which excels in long-term and large-scale predictions. Both models are applied to process the low-frequency and high-frequency components derived from VMD decomposition, respectively. The final prediction results were obtained by integrating the outputs from both models.

The Fig. below illustrates the results of partial VMD on the training dataset 1. In VMD decomposition, the value of $\boldsymbol{K}$ is primarily determined through experience or repeated experimentation. In case 2, $\boldsymbol{K}$ is determined based on the similarity of the center frequencies of each IMF after VMD decomposition. To define the similarity of IMFs, the following rule is established: for two adjacent IMFs, if the center frequency of the latter is less than 50 % higher than that of the former, then these two IMFs are considered similar. The center frequencies of each IMF under different $\boldsymbol{K}$ values for dataset 1 after VMD decomposition are shown in Table 5. When $\boldsymbol{K}$=2, 3, 4, 5, for any adjacent IMFs, the center frequency of the latter is at least 1.5 times that of the former, indicating effective decomposition. However, when $\boldsymbol{K}$=6, the center frequency of IMF 5 does not exceed 50 % of that of IMF 4, suggesting that the VMD has resulted in over-decomposition at this point. Therefore, the number of VMD decomposition modes is determined to be 5 Figs. 9-14.

The prediction results of the MC-VMD-ARIMA-LSTM model are shown in Fig. 15. Compared to the MC-LSTM prediction model, the curve generated by the MC-VMD-ARIMA-LSTM model exhibits fewer fluctuations influenced by noise and more closely follows the trend of the actual value curve. MC-VMD-ARIMA-LSTM model reduces high-frequency fluctuations by 62 % compared to MC-LSTM, minimizing noise-induced deviations during rapid load transitions. The enhanced performance stems from the VMD-ARIMA-LSTM synergy.

To more intuitively demonstrate the superiority of the MC-VMD-ARIMA-LSTM prediction results, Fig. 16 presents a comprehensive quantitative comparison of prediction accuracy between the MC-VMD-ARIMA-LSTM and MC-LSTM model. The histograms comparing the MSE, RMSE, MAPE, and R^2 of the prediction results are shown below. The RMSE, MAE, and MAPE of the prediction results decreased by 38 %, 21.6 %, and 54 %, R^2 of the MC-VMD-ARIMA-LSTM algorithm is 65 % higher than that of MC-LSTM. All of these results indicate a significant reduction in errors and confirm our algorithm's suitability for actual distribution system forecasting applications.

5. Conclusion

This paper proposes an alternating minimization matrix completion method based on constrained low-rank matrices to analyze and complete measurement data from a single section of the distribution network under low-observability conditions. On this basis, combining ARIMA and LSTM algorithms, we perform completion under time-scale missing conditions in distribution network measurement data. To better adapt the algorithm to the features of distribution network measurement data and to improve the completion effect, we further optimize the algorithm and propose the MC-VMD-ARIMA-LSTM algorithm. The matrix completion and MC-ARIMA algorithm proposed in this paper are tested on the IEEE 33-bus standard distribution network, and the MC-LSTM and MC-VMD-ARIMA-LSTM algorithms are verified by using the measured data of the actual distribution system.

The proposed algorithms demonstrate robust performance across multiple validation scenarios. The matrix completion algorithm on the IEEE 33-bus standard distribution network demonstrated high accuracy and showed good matching with ARIMA in ultra-short-term forecasting, the estimation error (MAPE) remained below 10 %. In practical distribution systems with larger time scales and datasets, the MC-LSTM algorithm demonstrated the reliability of the matrix completion algorithm in handling state estimation for actual low-observation condition (around 40 %-50 % data loss). To further mitigate the impact of measurement noise and some inevitable completion errors on overall prediction accuracy, the completed dataset was processed through VMD, then ARIMA and LSTM were used to predict low-frequency and high-frequency signals, respectively. This ultimately yielded more accurate predicted measurement data, reducing estimation error (MAPE) by 54 % compared to single prediction model.

The increasing integration of distributed energy resources and dynamic loads has heightened the demand for monitoring distribution networks. In the future, it will be necessary to monitor the status of distributed energy resources and dynamic loads across a broader range of time scales, extending from day-ahead forecasting to the frequency of real-time data uploads to data acquisition systems. The algorithms in this paper help to solve critical challenges in modern distribution networks with high DER penetration and provide a robust solution for enhancing operational visibility in low-observability environments.

Future research will focus on the following aspects:

(1) Focusing more intently on the observability of spatiotemporal measurement data within low-voltage distribution networks. These networks are intimately connected to end-users, and any line fluctuations caused by the integration of distributed energy resources and new types of loads can directly impact consumers, posing potential safety hazards.
(2) Incorporating specific load profiles into the matrix completion algorithm's considerations is essential. For instance, different types of loads such as schools, industrial parks, and commercial buildings each possess unique electricity consumption characteristics. Profiling these users can lead to more accurate completion models, thereby enhancing the precision and specificity of data completion.

Declaration of Generative AI technologies in the writing process

During the preparation of this work the author(s) used DeepSeek in order to improve readability. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

CRediT authorship contribution statement

Xiaodan Yu: Formal analysis. **Ruijia Jiang:** Methodology, Writing – original draft. **Xiaolong Jin:** Validation, Conceptualization. **Hongjie Jia:** Project administration. **Yunfei Mu:** Supervision. **Wei Wei:** Resources. **Wanxin Tang:** Visualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Xiaolong Jin reports financial support was provided by the project of National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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